Spaceflight alters the tomato phyllosphere and rhizosphere microbiome in a light-dependent manner: a multi-omics integration of ISS VEG-05

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Abstract

Spaceflight imposes unique stressors on plant-microbe interactions, but the coordinated responses of host transcriptomes and associated microbiomes remain poorly characterized. We analyzed tomato (*Solanum lycopersicum* cv. Red Robin) grown aboard the International Space Station (ISS) as part of the VEG-05 experiment, integrating 16S rRNA and ITS amplicon sequencing (OSD-766) with host transcriptomics (OSD-767) across two tissues (leaf, adventitious root) and two light treatments (red-rich vs. blue-rich LED). DADA2 processing yielded 348 bacterial and 77 fungal ASVs from 142 and 141 samples respectively. Leaf bacterial community composition differed among flight and light groups (PERMANOVA $\boldsymbol{R^2 = 0.42}$ for the full flight × light model, $\boldsymbol{p = 0.001}$), and flight samples had higher bacterial alpha diversity (Observed ASVs: Flight 17.8–25.8 vs. Ground 10.0–15.2). The spaceflight transcriptional response was strongly light-dependent: under blue light, 4,716 differentially expressed genes (DEGs) were detected in leaves versus only 523 under red light. WGCNA identified a 169-gene adventitious root module correlated with fungal dysbiosis ($\boldsymbol{r = -0.85}$, $\boldsymbol{p_{adj} = 0.001}$) but not with flight status ($\boldsymbol{r = -0.38}$, $\boldsymbol{p_{adj} = 0.32}$), representing a microbe-driven host transcriptional signature independent of the direct spaceflight stimulus. MOFA+ integration across three omics layers confirmed a dominant flight-associated factor (48% transcriptome variance) [**TODO: the MOFA+ microbiome views were built from mismatched samples; re-run before comparing bacterial and fungal responses**]. The most abundant FAPROTAX-predicted bacterial functions were aerobic chemoheterotrophy, nitrogen fixation, dark hydrogen oxidation and methanol oxidation. These findings demonstrate that spaceflight restructures plant-associated microbiomes in a tissue- and light-specific manner, with implications for crop production systems in long-duration space missions.

Keywords: spaceflight, plant microbiome, multi-omics, VEG-05, tomato, WGCNA, MOFA+, dysbiosis

1 Introduction

Plant-associated microbiomes play critical roles in nutrient acquisition, pathogen protection, and stress tolerance [1]. In the context of spaceflight, these interactions take on added significance: crops grown in bioregenerative life support systems must maintain productive plant-microbe partnerships under conditions of microgravity, altered fluid dynamics, elevated radiation, and controlled-environment lighting **[TODO: citation needed]**. The VEG-05 experiment aboard the International Space Station (ISS) provided a unique opportunity to study these interactions in tomato (*Solanum lycopersicum*), a candidate crop for long-duration missions [2, 3].

Spaceflight has been shown to alter microbial community composition, gene expression, and virulence in pure culture systems **[TODO: citation needed]**. However, the coordinated response of host plants and their associated microbiomes to the spaceflight environment remains largely unexplored. Multi-omics integration approaches such as MOFA+ (Multi-Omics Factor Analysis) [4] and WGCNA (Weighted Gene Co-expression Network Analysis) [5] offer powerful frameworks for identifying coordinated variation across biological layers, but have not been applied to spaceflight plant-microbiome studies.

Here we present an integrated analysis of the VEG-05 tomato microbiome (16S rRNA and ITS amplicon sequencing, OSD-766) and host transcriptome (RNA-seq, OSD-767) (Fig. 1). We characterize community health through diversity metrics and a dysbiosis index, identify co-expression modules associated with microbial changes, and infer the directionality of host-microbe interactions using a framework that distinguishes host-driven, microbe-driven, and co-regulated responses.

2 Methods

2.1 Data acquisition and metadata

Raw sequencing data and metadata were obtained from NASA's Open Science Data Repository (OSDR): OSD-766 (microbiome amplicon sequencing; [2]) and OSD-767 (host RNA-seq; [3]). The VEG-05 experiment grew tomato cv. Red Robin aboard the ISS under red-rich (90% red:10% blue) or blue-rich (50% red:50% blue) LED lighting, with ground controls at Kennedy Space Center. Metadata were parsed from ISA-Tab format, and a crosswalk table linking microbiome and RNA-seq samples by plant ID, flight status, light treatment, and tissue was constructed.

2.2 Microbiome amplicon processing

16S rRNA (V4 region, 515F/806R primers) and ITS2 (ITS3F/ITS4 primers) reads were processed using DADA2 v1.34.0 [6] in R. For 16S, primers were already removed;

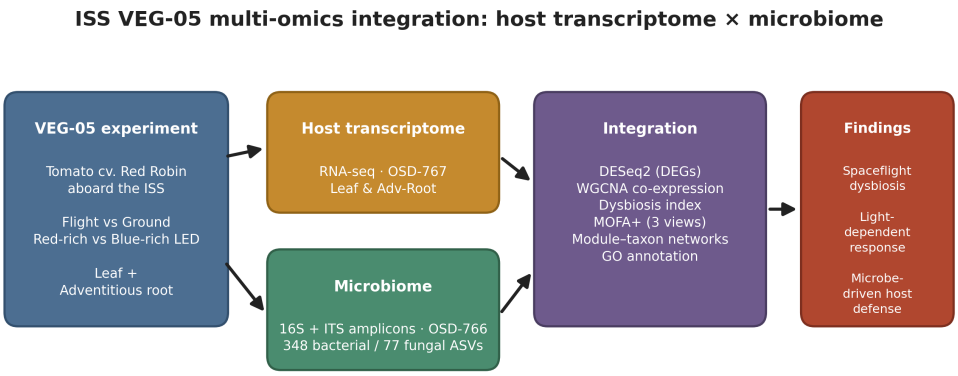


Fig. 1 Study design and analysis overview. The ISS VEG-05 experiment (tomato cv. Red Robin; Flight vs Ground; red-rich vs blue-rich LED; leaf and adventitious root) feeds two data streams — host transcriptome (OSD-767) and microbiome (OSD-766) — combined through DESeq2, WGCNA, a dysbiosis index, MOFA+, and module–taxon networks.

reads were truncated at 240 bp (R1) and 160 bp (R2) with maxEE = 2, length-filtered to 200–260 bp, and pooled for denoising. Taxonomy was assigned against SILVA 138.1 [7]. Chloroplast and mitochondrial ASVs were removed prior to community analysis. For ITS, primers were removed with cutadapt [8]; R2 reads were truncated at 180 bp due to quality degradation; taxonomy was assigned against UNITE v7 (October 2017) [9].

2.3 Differential expression analysis

RNA-seq count matrices were analyzed with DESeq2 [10] separately for leaf (21 samples) and adventitious root (15 samples; Table S1). The design model included flight status, light treatment, and their interaction. Log-fold changes were shrunk with apeglm [11] for main effects and ashr [12] for subset contrasts. DEGs were defined as $p_{\text{adj}} < 0.05$ and $|\log2\text{FC}| \geq 1$.

2.4 Community health metrics

Alpha diversity (Observed ASVs, Shannon index) was computed on unfiltered data using phyloseq [13]. Beta diversity was assessed via Bray-Curtis dissimilarity and PERMANOVA (999 permutations) using vegan [14]. A dysbiosis index was calculated for each flight sample as its mean Bray-Curtis distance to all ground samples of the same compartment (both light treatments), divided by the mean pairwise distance among those ground samples. Differential abundance testing used ALDEx2 [15] with 128 Monte Carlo instances, requiring both a Benjamini–Hochberg-adjusted Welch $p < 0.05$ and $|\text{effect size}| > 1$; positive effects are higher in flight.

2.5 WGCNA co-expression networks

WGCNA [5] was applied to variance-stabilized transcriptome data for each tissue separately. Soft-thresholding powers were selected to maximize scale-free fit (Leaf: power = 20, $R^2 = 0.72$; Adv-Root: power = 18, $R^2 = 0.69$). Module eigengenes were correlated (Pearson) with flight status, light treatment, and 16S/ITS dysbiosis indices. Hub genes were the 10 genes with the highest module membership (k_{ME}) in each module (all $k_{\mathrm{ME}} \geq 0.86$).

2.6 MOFA+ multi-omics integration

MOFA2 [4] was trained on 21 leaf samples with matched data across three views: transcriptome (2,000 most variable genes), 16S (348 ASVs), and ITS (77 ASVs). The model was trained with 5 factors and default priors. Factor-trait correlations were computed using Spearman's rank correlation with BH correction.

2.7 Correlation networks and directionality inference

Bipartite module-taxon networks were constructed by Spearman correlation between WGCNA module eigengenes and ASV relative abundances (prevalence filter $> 20\%$, BH correction). GO enrichment of WGCNA modules was performed using clusterProfiler's enricher function [16] with tomato GO annotations from Ensembl Plants (18,479 annotated genes). Directionality was classified as: host-driven (correlated with flight/light but not dysbiosis), microbe-driven (correlated with dysbiosis but not flight/light), or co-regulated (correlated with both).

2.8 Functional prediction

Bacterial functional profiles were predicted using FAPROTAX v1.2 [17] with a custom Python implementation of the collapse algorithm. Fungal ecological guilds were assigned using genus-level lookup tables based on FUNGuild [18] categories.

3 Results

3.1 Spaceflight reshapes leaf bacterial communities

DADA2 processing of 16S rRNA reads yielded 544 ASVs from 142 samples (3,558,814 non-chimeric reads, 89.8% retention). After removing chloroplast (38.9% of reads) and mitochondrial (21.5%) sequences, 348 bacterial ASVs remained. ITS processing yielded 77 fungal ASVs from 141 samples (160,316 reads, 15.8% retention).

Bacterial community composition differed among flight and light groups in leaves (PERMANOVA, full flight $\times$ light model: $R^2 = 0.42$, $F = 4.02$, $p = 0.001$, $n = 21$) and adventitious roots ($R^2 = 0.33$, $F = 2.93$, $p = 0.001$, $n = 15$; Table S4). These R^2 values are for the whole model, not for spaceflight alone **[TODO: term-wise PERMANOVA]**. In leaves, bacterial alpha diversity was higher in flight samples: Observed ASVs ranged from 17.8–25.8 in flight versus 10.0–15.2 in ground controls (Fig. 2; Table S3). The Shannon index showed a similar pattern, with flight blue-light samples reaching 2.18 versus 0.80 in ground blue-light controls.

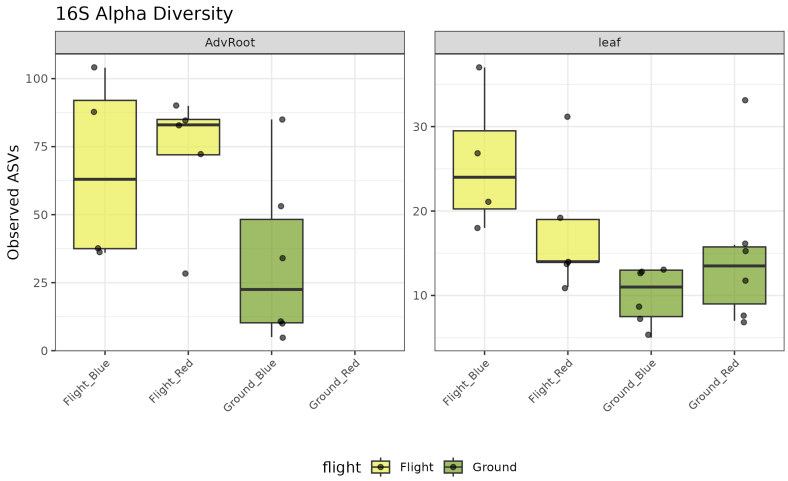


Fig. 2 Bacterial (16S) alpha diversity (Observed ASVs) by flight status and light treatment; flight leaves show higher diversity.

The dysbiosis index quantified community displacement relative to ground controls. Leaf bacterial communities showed a 2.06-fold increase in dysbiosis during flight (2.06 ± 0.99 vs. 1.00 ± 0.47; Fig. 3). Fungal communities showed even greater displacement (ITS leaf: 3.64 ± 2.78 vs. 1.00 ± 0.24; Fig. 3), though the PERMANOVA model was not significant (model $R^2 = 0.45$, $p = 0.054$). Adventitious root fungal community composition also differed among groups (model $R^2 = 0.65$, $F = 8.19$, $p = 0.001$).

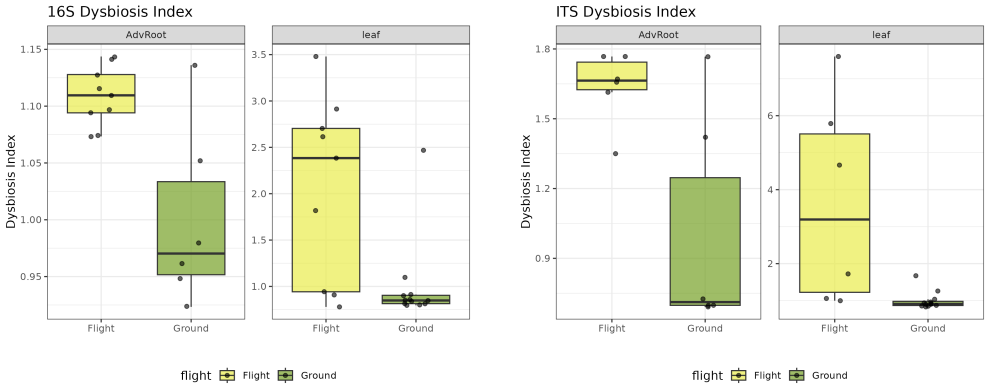


Fig. 3 Dysbiosis index (mean Bray-Curtis distance to ground samples, relative to ground-ground distance). Left (16S): bacterial dysbiosis, elevated under flight. Right (ITS): fungal dysbiosis, largest displacement in flight leaves.

ALDEx2 identified few taxa with significant differential abundance under the conservative dual criterion (BH-adjusted $p < 0.05$ and $|\text{effect}| > 1$), all of them more abundant in flight. In leaf 16S, *Pantoea* (ASV024) was enriched in flight (effect

= +1.10). In ITS, *Penicillium citrinum* (ASV003) was enriched in flight leaves (effect = +1.82). In adventitious roots, *Fusarium solani* (ITS ASV006; effect = +1.82) and three bacterial taxa (16S ASV004, *Burkholderia-Caballeronia-Paraburkholderia*, +1.58; ASV021, Rhizobiaceae, +3.25; ASV022, *Dyadobacter*, +2.22) were enriched in flight.

3.2 Spaceflight transcriptional response is light-dependent

DESeq2 analysis revealed a pronounced interaction between spaceflight and light quality (Table S2). In leaves, the flight term of the flight × light model, which estimates the flight effect under red-rich light (the reference level), gave 757 DEGs (404 up, 353 down; Fig. 5). Separate models within each light treatment gave 4,716 DEGs under blue-rich light and 523 under red-rich light (Fig. 4); the two red-rich estimates differ because they come from different models. The formal interaction term identified 3,189 DEGs, confirming that the transcriptional response to spaceflight is strongly modulated by light quality.

In adventitious roots, the flight term (red-rich reference level) was predominantly upregulation (896 DEGs: 783 up, 113 down, 87% upregulated; Fig. 5), suggesting activation of stress response pathways. The interaction was minimal (6 DEGs after shrinkage), indicating that the root transcriptional response to spaceflight is less light-dependent than the leaf response.

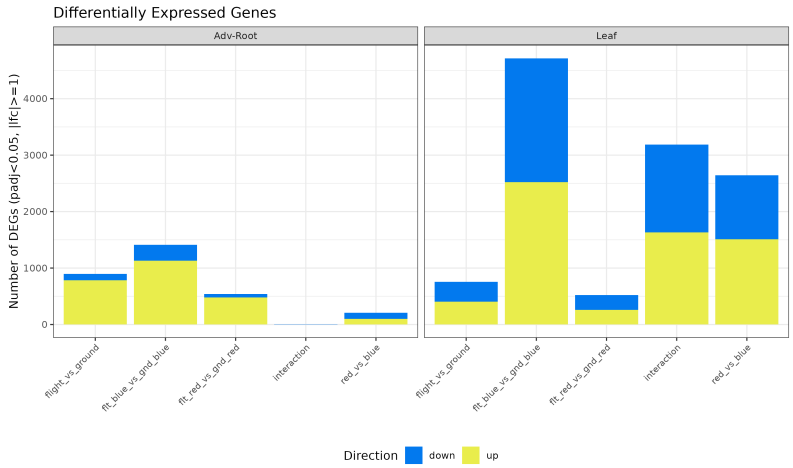


Fig. 4 DEG summary across tissue × contrast; the leaf blue-light flight response (4,716 DEGs) dwarfs red (523).

3.3 WGCNA identifies a microbe-driven host transcriptional module

WGCNA of the leaf transcriptome identified 12 modules plus unassigned genes (power = 20; Fig. 6), with the turquoise module (n = 2,037 genes) showing the strongest flight

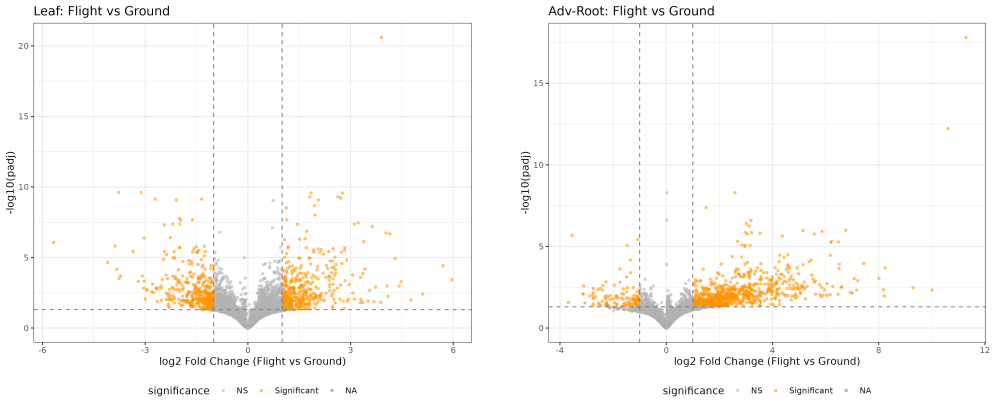


Fig. 5 Flight-vs-Ground volcano plots for the flight term of the flight × light model, i.e. the flight effect under red-rich light (the reference level). Left (3*B*): leaf. Right (3*C*): adventitious root (predominantly upregulation).

correlation ($r = +0.73$, $p_{\text{adj}} = 0.001$) and the blue module ($n = 1{,}012$) the strongest anti-correlation ($r = -0.79$, $p_{\text{adj}} = 3 \times 10^{-4}$). Four leaf modules were classified as co-regulated (correlated with both flight and dysbiosis; Table S5) [**TODO: Table S5 was built from an earlier correlation table; re-derive**].

In adventitious roots, WGCNA identified 16 modules plus unassigned genes (power = 18). The key finding was the black module ($n = 169$ genes), strongly correlated with ITS dysbiosis ($r = -0.85$, $p_{\text{adj}} = 0.001$) but not with flight status ($r = -0.38$, $p_{\text{adj}} = 0.32$; Fig. 6). This module was classified as microbe-driven, representing a host transcriptional response to fungal community changes independent of the direct spaceflight stimulus. GO enrichment of the black module itself was limited — its only significantly enriched term was the cellular component "nucleus" — so it is defined here by its microbe-driven trait association; the adventitious-root oxidative-stress program was instead concentrated in the larger turquoise module (see below).

GO enrichment of all modules revealed biologically coherent signatures (Table S10). In leaves, the black module was enriched for photosynthesis and photosystem genes ($p_{\text{adj}} = 4.4 \times 10^{-25}$), while the red module was enriched for RNA binding and protein binding. In adventitious roots, the turquoise module (the largest, $n = 1{,}768$) was enriched for oxidative stress response (peroxidase activity, $p_{\text{adj}} = 4.2 \times 10^{-9}$; hydrogen peroxide catabolism, $p_{\text{adj}} = 1.1 \times 10^{-8}$; glutathione transferase activity, $p_{\text{adj}} = 3.9 \times 10^{-7}$), consistent with spaceflight-induced oxidative stress.

3.4 Module functional annotation (GO)

To make the WGCNA module colours interpretable, each module was annotated with its dominant biology from per-module Gene Ontology enrichment (Table 1). These human-readable labels are shown alongside the module colours on the axes of Figs 6 and 8. Modules with no significant enrichment are marked accordingly.

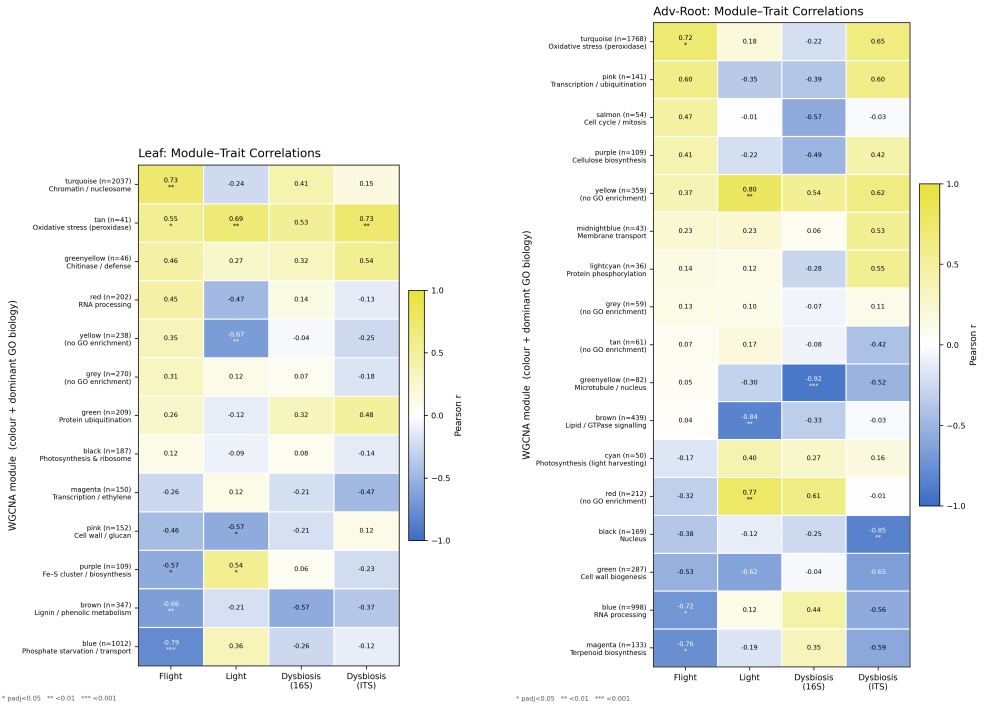


Fig. 6 WGCNA module–trait correlations (flight, light, 16S/ITS dysbiosis). Left: leaf. Right: adventitious root — the black (nucleus) module tracks ITS dysbiosis ($r = -0.85$) but not flight; the turquoise (oxidative-stress/peroxidase) module is the largest flight-correlated module. Each module is labelled with its dominant GO biology (Table 1).

Table 1: GO-based functional annotation of WGCNA modules.

Tissue / Module	GO biology (dominant)	Representative GO terms
Adv-Root / black	Nucleus	nucleus
Adv-Root / blue	RNA processing	protein binding; RNA binding; RNA modification
Adv-Root / brown	Lipid / GTPase signalling	ATP-dependent diacylglycerol kinase activity; phospholipase C-activati...
Adv-Root / cyan	Photosynthesis (light harvesting)	photosynthesis, light harvesting; photosynthesis; photosystem I
Adv-Root / green	Cell wall biogenesis	xyloglucan metabolic process; cell wall; xyloglucan:xyloglucosyl trans...
Adv-Root / greenyellow	Microtubule / nucleus	microtubule binding; ATP binding; nucleus
Adv-Root / lightcyan	Protein phosphorylation	protein phosphorylation; protein kinase activity; ATP binding

Table 1 — continued

Tissue / Module	GO biology (dominant)	Representative GO terms
Adv-Root / magenta	Terpenoid biosynthesis	acyltransferase activity; terpene synthase activity; terpenoid biosynt...
Adv-Root / midnightblue	Membrane transport	transmembrane transporter activity; transmembrane transport; membrane
Adv-Root / pink	Transcription / ubiquitination	DNA-binding transcription factor activity; ubiquitin protein ligase ac...
Adv-Root / purple	Cellulose biosynthesis	cellulose synthase (UDP-forming) activity; cellulose biosynthetic proc...
Adv-Root / salmon	Cell cycle / mitosis	cyclin-dependent protein serine/threonine kinase regulator activity; m...
Adv-Root / turquoise	Oxidative stress (peroxidase)	peroxidase activity; hydrogen peroxide catabolic process; heme binding
Leaf / black	Photosynthesis & ribosome	photosynthesis; ribosome; photosystem I
Leaf / blue	Phosphate starvation / transport	DNA-binding transcription factor activity; cellular response to phosph...
Leaf / brown	Lignin / phenolic metabolism	heme binding; endopeptidase inhibitor activity; cation binding
Leaf / green	Protein ubiquitination	protein ubiquitination; ubiquitin-protein transferase activity; cell s...
Leaf / greenyellow	Chitinase / defense	chitinase activity; chitin catabolic process; chitin binding
Leaf / magenta	Transcription / ethylene	regulation of DNA-templated transcription; ethylene-activated signalin...
Leaf / pink	Cell wall / glucan	cell wall; xyloglucan:xyloglucosyl transferase activity; glucan metabo...
Leaf / purple	Fe–S cluster / biosynthesis	iron-sulfur cluster binding; biosynthetic process
Leaf / red	RNA processing	RNA binding; protein binding; RNA modification
Leaf / tan	Oxidative stress (peroxidase)	hydrogen peroxide catabolic process; peroxidase activity; response to...
Leaf / turquoise	Chromatin / nucleosome	nucleosome; structural constituent of chromatin; microtubule binding

3.5 MOFA+ confirms dominant flight factor and cross-omics coordination

MOFA+ integration of 21 matched leaf samples across three omics views (transcriptome: 2,000 genes; 16S: 348 ASVs; ITS: 77 ASVs) identified 5 factors [**TODO: the 16S and ITS views were built from mismatched samples (tissue not matched, columns misordered); the microbiome variance fractions and Factor 2–3 associations need re-running**]. Factor 1 was the dominant signal, explaining 48.0% of transcriptome variance, 12.0% of 16S variance, and 3.0% of ITS variance (Fig. 7). Factor 1 was significantly correlated with flight status (Spearman $\rho = -0.76$, $p_{\text{adj}} = 0.001$; Fig. 7), confirming that spaceflight is the primary axis of coordinated variation across omics layers. Factor 2 trended with 16S dysbiosis ($\rho = -0.52$, $p = 0.018$, $p_{\text{adj}} = 0.089$), while Factor 3 trended with both light treatment ($\rho = 0.52$) and ITS dysbiosis ($\rho = 0.50$). The transcriptome contributed the highest-weighted features to all factors, but microbial features were consistently present in the top weights (Fig. S6).

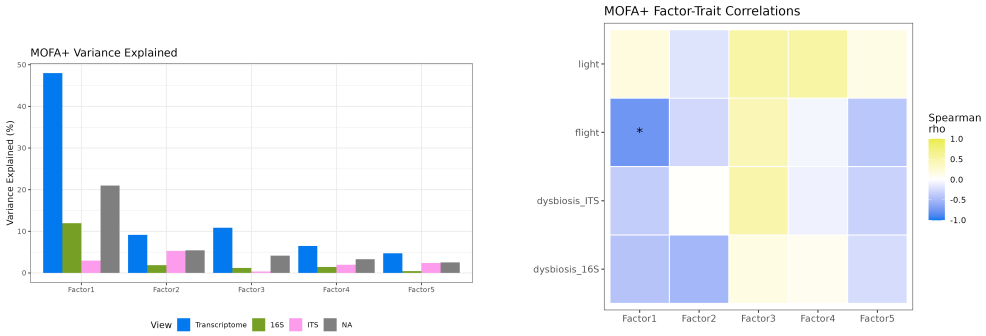


Fig. 7 MOFA+ integration. Left (5*A*): variance explained per factor per view. Right (5*B*): factor–trait correlations; Factor 1 captures flight ($\rho = -0.76$).

3.6 Module-taxon correlations link host modules to specific bacteria

Bipartite correlation networks identified 29 significant module-taxon relationships (BH $p_{\text{adj}} < 0.05$), predominantly in leaf 16S (28 of 29; Fig. 8; Table S9). The flight-associated turquoise module positively correlated with *Methylobacterium-Methylorubrum* ($\rho = +0.76$, $p_{\text{adj}} = 0.003$), *Burkholderia-Caballeronia-Paraburkholderia* ($\rho = +0.74$, $p_{\text{adj}} = 0.003$), and *Azospirillum* ($\rho = +0.68$, $p_{\text{adj}} = 0.012$) — all genera associated with plant growth promotion and nitrogen fixation. The anti-correlated blue module showed the inverse pattern. The brown module correlated negatively with *Pantoea* ($\rho = -0.79$) and *Paenibacillus* ($\rho = -0.77$).

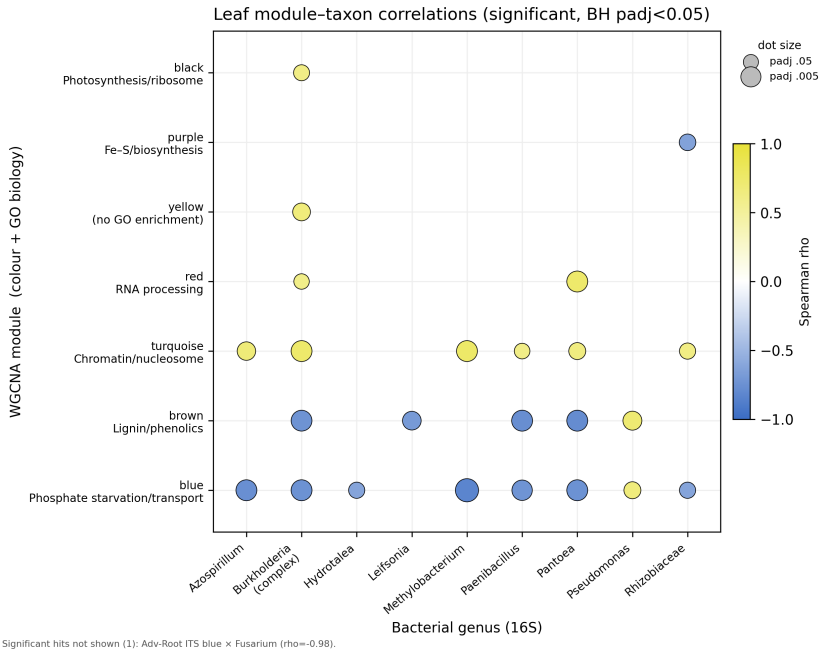


Fig. 8 Leaf module–taxon correlations (BH $p_{\mathrm{adj}} < 0.05$). The flight-associated turquoise (chromatin/nucleosome) module positively tracks *Methylobacterium*, *Burkholderia*, and *Azospirillum*, while the blue (phosphate-starvation) module shows the inverse.

3.7 Predicted bacterial functions

FAPROTAX assigned bacterial ASVs to 26 functional categories (Table S6). By read count the largest were aerobic chemoheterotrophy (787,792 reads, 185 taxa), nitrogen fixation (164,614 reads, 12 taxa), dark hydrogen oxidation (103,694 reads, 20 taxa), methanol oxidation (96,687 reads, 15 taxa) and methanotrophy (96,514 reads, 15 taxa) (Fig. 9). The methanol oxidation and methanotrophy assignments were primarily *Methylobacterium-Methylorubrum*, the same genus showing the strongest positive correlation with the flight-associated WGCNA module [**TODO: the FAPROTAX collapse misparses three database lines, moving 31 ASVs into spurious categories, and matches taxon names as substrings (*Methylobacterium* is a methylotroph, not a methanotroph); re-run with a corrected parser**]. Fungal guild assignment (manual genus-level classification) identified saprotrophs (11 ASVs, dominated by *Penicillium* and *Aspergillus*) and plant pathogens (2 ASVs: *Fusarium* spp.) as the main guilds (Table S7).

4 Discussion

This study presents the first integrated multi-omics analysis of plant-microbiome interactions in the ISS VEG-05 experiment. Our findings reveal three key insights:

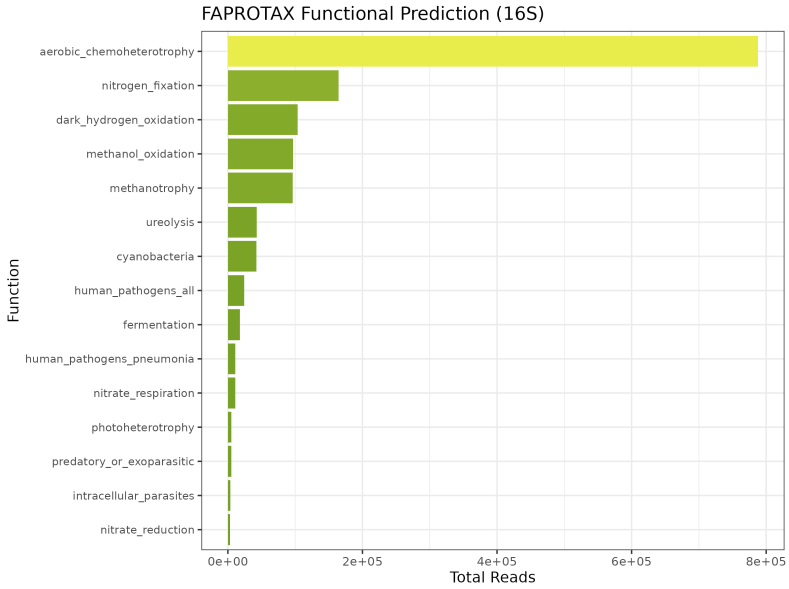


Fig. 9 FAPROTAX predicted bacterial functions (aerobic chemoheterotrophy, N fixation, methanotrophy, methanol oxidation).

(1) spaceflight restructures leaf bacterial communities more than fungal communities, with increased alpha diversity in flight; (2) the host transcriptional response to spaceflight is strongly light-dependent, particularly in leaves; and (3) a subset of host genes responds to microbial community changes independently of the direct spaceflight stimulus, providing evidence for microbe-driven host regulation.

The increased bacterial alpha diversity in flight leaves was initially counterintuitive, as spaceflight is often associated with reduced microbial diversity in human microbiome studies [19]. However, in the phyllosphere, the reduced gravity environment may alter leaf surface properties and nutrient availability, potentially creating more ecological niches. The strong correlation between the flight-associated WGCNA module and plant growth-promoting bacteria (*Methylobacterium*, *Burkholderia*, *Azospirillum*) suggests that spaceflight may favor these beneficial taxa, though the directionality of this association requires further investigation.

The light-dependent transcriptional response is a novel finding with practical implications for space crop production. The 9-fold difference in DEG counts between blue and red light conditions (4,716 vs. 523) suggests that blue light may amplify the plant's perception of or response to the spaceflight environment. Blue light is known to activate cryptochrome photoreceptors and downstream stress signaling pathways [20], which may interact with spaceflight-induced oxidative stress.

The identification of a microbe-driven WGCNA module in adventitious roots (black module, 169 genes) is perhaps the most significant finding. This module's strong correlation with fungal dysbiosis ($r = -0.85$) but not with flight status ($r = -0.38$, $p_{\text{adj}} = 0.32$) indicates that the host transcriptional response in this module is driven

by microbial community changes rather than directly by the spaceflight environment. Although the black module's own GO enrichment was limited, the broader adventitious-root oxidative-stress program — concentrated in the larger turquoise module — together with the black module's tight coupling to fungal dysbiosis suggests that fungal community restructuring may trigger host defense and redox responses, potentially through oxidative burst signaling.

4.1 Relationship to a cross-mission salicylic-acid integration

These microbiome findings dovetail with a companion cross-mission analysis that integrated the same OSD-767 transcriptome with an Advanced Plant Habitat experiment manipulating salicylic-acid (SA) signalling (wild-type MoneyMaker versus the SA-deficient *NahG* line). That analysis concluded that the spaceflight transcriptome resembles an SA-deficient, biotic-defense-primed state. Our results offer a proximal explanation: spaceflight-induced dysbiosis — and the microbe-driven, flight-independent host module identified here — could be the stimulus that elicits that defense-primed signature, with SA acting as the gating regulator of plant biotic immunity. Both analyses independently recover the blue-light amplification of the spaceflight response despite using different pipelines, and this rationalises why gene-level spaceflight signatures replicate poorly across missions while the biotic-defense *pattern* is conserved. A full comparison is provided in `COMPARISON_with_SA_integration.md`.

4.2 Limitations

Several limitations should be noted. First, sample sizes are small ($n = 21$ leaf, $n = 15$ root for RNA-seq; $n = 1$–6 per flight $\times$ light group for microbiome), limiting statistical power. Second, 16S leaf samples had very low bacterial read counts after host DNA removal (median 89 reads), precluding rarefaction and requiring reliance on relative abundance metrics. Third, the ITS dataset had uneven group representation, with some flight groups containing only 1–2 samples. Fourth, MOFA+ warned that Factor 1 may partially capture technical variation (detection rate differences), though the strong biological signal supports its interpretation as a flight factor. Fifth, FAPROTAX functional predictions are taxonomy-based and should be validated with metagenomic or metatranscriptomic data. Sixth, the FUNGuild analysis used manual genus-level assignment rather than the full database, reducing assignment confidence. Finally, this study is correlational; causal relationships between host transcription and microbiome changes require experimental validation, ideally in simulated microgravity conditions with microbiome manipulation.

Data and code availability. All analysis code is available at https://github.com/dr-richard-barker/veg05-integrated-omics. Raw data are available from NASA OSDR (OSD-766, OSD-767).

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